

CyberDefense Pro 2.0

2.1.6 Open Web Application Security Project (OWASP)

The Open Web Application Security Project (OWASP) is a nonprofit foundation that works to improve the security of web applications and services. It is an international organization that provides unbiased, practical information about application security. The OWASP provides tools, documents, and other resources to help people build more secure software.

This lesson covers the following topics:

- Open Web Application Security Project (OWASP)
- OWASP Projects

Open Web Application Security Project (OWASP)

The Open Web Application Security Project (OWASP) is a not-for-profit organization dedicated to improving the security of software and applications online. OWASP was founded in 2001, and its mission is to provide free, open-source tools and resources to help developers and organizations create more secure applications and services. OWASP promotes awareness of web application security issues, develops resources to educate developers and users, and offers various testing tools to help organizations identify and fix security vulnerabilities. OWASP's resources are used by developers, security professionals, and organizations worldwide to build and maintain secure applications.

OWASP's commitment to open-source software and information sharing has made it an invaluable resource for the security community.

OWASP's mission is to raise awareness of the risks of building insecure software. The goals of OWASP are to do the following:

- Create awareness of risks and vulnerabilities in software applications.
- Create and promote tools and resources for developers to create more secure applications.
- Provide a forum for individuals to contribute and collaborate on security initiatives through open-source software.

OWASP's resources include free web application security tools, training, and other resources designed to help organizations identify and fix application security vulnerabilities. OWASP's

community-driven approach to software security provides organizations with various resources, including software tools, guides, training, and research. These resources help organizations improve software security, prioritize and plan security initiatives, and meet compliance standards. OWASP's commitment to open-source software and information sharing has made it an invaluable resource for the security community. OWASP's resources are free, easy to use, and available to everyone, making them an ideal solution for organizations of all sizes and industries.

The most common web application vulnerabilities are cross-site scripting (XSS), SQL injection, path traversal, broken authentication and authorization, and insecure direct object references. OWASP offers a variety of testing tools, such as Zed Attack Proxy, to help organizations identify and fix security vulnerabilities related to all of these issues. OWASP's resources also include a variety of guides, training, research, and other resources to help organizations improve software security. For example, the OWASP Testing Guide describes testing tools, including specific instructions for finding and identifying vulnerabilities in web applications. The organization's tools and resources are free, easy to use, and available online through printed guidebooks and other materials. Anyone can contribute to OWASP's resources, including researchers and developers at organizations large and small. Contributors can suggest edits and additions to existing content and submit new content. It is also possible to translate OWASP content into different languages to make it accessible to a broader audience.

Organizations use OWASP's resources and tools to create more secure applications and services. They can also benefit from contributing their resources and tools by becoming part of the OWASP community. Organizations can use OWASP's resources to improve their software security, meet compliance standards, and demonstrate their commitment to data security. Contributing to OWASP's resources allows organizations to share best practices, make their work accessible to a broader audience, and become part of the global security community.

OWASP Projects

The OWASP Top 10

The OWASP Top 10 guides describe and prioritize serious web application security vulnerabilities. The OWASP Top 10 represents a consensus view of the most pressing and critical web application security issues based on various sources, including real-world security data, research, and experience.



Image courtesy of OWASP Foundation, Inc.

▼ Description

The image is the logo for OWASP Top 10, a well-known cybersecurity standard. At the top left, there is a dark blue outline of a circle containing the silhouette of a wasp, which represents the OWASP organization (Open Web Application Security Project). To the right of the wasp symbol, the word "OWASP" is written in capital letters. Below this, the phrase "TOP 10" is displayed in very large, bold blue characters, with "TOP" and "10" in different shades of blue. Underline styling appears beneath the "10". The image visually emphasizes the importance of the "Top 10" list, which refers to the ten most critical web application security risks identified by OWASP.

OWASP Top 10 is an incredibly important body of work that is frequently referenced. The OWASP Top 10 is periodically updated and available at <https://owasp.org/Top10/>.

As of this writing, the most recent revision to the OWASP Top 10 occurred in 2021 and describes the following vulnerabilities (in order of importance):

1. Broken Access Control
2. Cryptographic Failures
3. Injection
4. Insecure Design
5. Security Misconfiguration
6. Vulnerable and Outdated Components
7. Identification and Authentication Failures
8. Software and Data Integrity Failures
9. Security Logging and Monitoring Failures
10. Server-Side Request Forgery

The OWASP ESAPI (Enterprise Security API)

The OWASP ESAPI (Enterprise Security API) is an application security standard. It's an application security framework that specifies an implementation approach for crucial security controls, including authentication and authorization, session management, cybersecurity hygiene, and secure coding practices.

OWASP ModSecurity

OWASP ModSecurity is a web application firewall that protects web applications against malicious traffic. It provides real-time detection of attacks and malicious user behavior that might otherwise go unnoticed and unhandled by standard security controls.

OWASP Parse Open

OWASP Parse Open is used to parse and transform structured content, including data feeds, structured documents, and data-heavy web pages. It is a free, open-source parsing and data extraction tool that makes extracting structured data from unstructured and semi-structured data sources easy.

The Open Crypto Audit Project (OCAP)

The Open Crypto Audit Project (OCAP) was established to help organizations understand the security of their systems while using cryptography to protect their data and assets. OCAP provides resources to help organizations identify and address risks related to cryptography, including code reviews, security reviews, and other tools and resources.

Copyright © CompTIA, Inc. All rights reserved.